

FOOTHILLS, NC, USA

WMO: 723148

Lat: 35.821N

Lon: 81.611W

Elev: 1270

StdP: 14.03

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 63859

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	20.9	-4.1	4.6	33.4	0.8	6.0	33.7	20.0	46.6	18.0	42.6	2.1	300	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	93.0	71.1	90.7	71.1	88.5	70.9	76.5	84.3	75.3	83.4	74.4	82.2	5.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.7	136.3	78.8	73.2	129.1	77.7	72.4	125.9	77.0	40.7	84.9	39.8	82.4	38.7	82.1	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.8	10.7	DB	7.1	96.0	6.4	2.0	2.5	97.4	-1.3	98.5	-4.9	99.6	-9.5	101.0
			WB	6.3	78.3	5.9	1.5	2.1	79.4	-1.4	80.3	-4.7	81.1	-9.0	82.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	39.2	41.8	49.7	58.8	66.4	73.9	76.8	75.7	69.4	58.3	48.6	42.4	(6)
	DBStd	15.16	10.03	8.48	9.06	7.75	6.54	4.31	3.85	4.15	5.61	7.77	7.80	8.11	(7)
	HDD50	1144	358	253	121	16	0	0	0	0	0	18	116	262	(8)
	HDD65	3660	801	650	477	212	67	2	0	1	25	232	492	701	(9)
	CDD50	4251	23	24	113	280	508	716	832	796	581	276	75	27	(10)
	CDD65	1288	0	1	3	26	110	268	367	332	155	25	1	0	(11)
	CDH74	11987	8	6	76	474	1161	2543	3416	2800	1236	237	29	1	(12)
	CDH80	4590	2	0	6	110	354	1000	1524	1159	396	37	2	0	(13)
(14) Wind	WSAvg	3.0	3.7	3.9	4.3	4.3	3.1	2.5	2.1	1.9	2.3	2.7	2.7	2.7	(14)
(15) Precipitation	PrecAvg	49.5	4.1	4.1	4.6	3.6	4.2	4.7	4.3	4.1	4.4	3.9	3.7	3.7	(15)
	PrecMax	65.7	8.9	8.4	12.5	8.3	10.9	10.9	9.4	9.4	11.9	12.8	10.5	8.3	(16)
	PrecMin	32.0	0.3	0.2	1.1	0.2	0.8	0.6	0.5	0.7	0.1	0.0	1.0	0.1	(17)
	PrecStd	7.6	1.8	2.1	2.2	2.0	2.0	2.1	2.2	2.2	2.9	2.9	1.9	2.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.0	73.5	81.0	87.6	90.9	94.5	96.7	95.4	91.1	83.9	79.0	72.0	(19)
		MCWB	56.9	60.4	59.0	60.7	66.5	70.2	70.9	72.6	68.1	67.2	61.1	58.2	(20)
	2%	DB	64.4	67.7	75.3	83.0	87.5	91.3	93.0	92.6	88.2	80.7	72.4	65.6	(21)
		MCWB	54.4	56.4	57.2	60.5	66.1	71.1	71.9	71.7	69.1	63.8	58.0	55.2	(22)
	5%	DB	60.9	63.1	72.0	80.4	83.9	88.9	90.6	89.8	84.4	76.4	68.1	61.2	(23)
		MCWB	52.3	50.4	56.3	60.2	66.0	70.6	71.8	72.5	69.5	62.8	56.3	53.5	(24)
	10%	DB	55.3	57.4	66.3	75.4	81.1	86.0	88.3	86.4	81.4	72.5	63.9	56.1	(25)
		MCWB	47.2	46.7	53.4	58.5	65.9	69.8	72.0	71.5	68.3	61.5	54.7	49.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.1	63.7	63.4	67.8	73.4	76.2	77.9	78.5	75.8	71.9	66.5	64.4	(27)
		MCDB	66.5	70.8	73.1	76.0	81.2	84.0	85.2	85.8	78.9	77.4	69.8	67.7	(28)
	2%	WB	58.2	59.0	61.0	65.5	71.2	74.9	76.3	76.9	74.2	69.3	62.9	59.4	(29)
		MCDB	63.0	64.6	69.0	75.0	78.8	83.0	85.1	84.5	79.2	74.0	66.7	62.2	(30)
	5%	WB	53.7	53.5	59.2	63.5	69.9	73.5	75.2	75.5	72.5	66.9	60.1	54.8	(31)
		MCDB	58.6	59.4	67.7	73.2	77.9	82.6	84.5	83.3	78.6	72.5	65.8	58.3	(32)
	10%	WB	49.5	49.1	56.5	61.6	68.4	72.4	74.2	74.3	70.9	64.1	56.8	50.5	(33)
		MCDB	54.3	55.6	64.1	70.7	76.4	80.9	82.7	81.6	77.0	68.8	62.5	54.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	22.0	23.9	25.9	24.0	22.9	21.8	21.1	20.7	23.6	25.2	20.6	(35)
		MCDBR	27.3	30.5	30.8	32.7	29.6	27.7	26.3	26.0	26.3	28.4	31.2	27.7	(36)
		MCWBR	17.1	18.3	15.2	13.4	12.0	9.8	8.9	9.0	10.3	14.3	17.5	16.7	(37)
	5% WB	MCDBR	22.6	24.0	24.5	24.2	22.9	22.1	21.4	20.1	19.3	20.2	21.1	20.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.299	0.307	0.338	0.388	0.436	0.466	0.509	0.502	0.399	0.355	0.314	0.299	(40)
		taud	2.532	2.502	2.455	2.314	2.200	2.151	2.059	2.057	2.333	2.449	2.529	2.562	(41)
		Ebn at Noon	288	298	296	284	271	261	249	248	272	276	279	281	(42)
	Edh at Noon		25	29	33	40	46	48	53	52	37	30	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		800	1005	1343	1673	1851	1969	1834	1702	1453	1188	897	688	(44)
	RadStd		51	113	90	130	146	139	129	83	132	141	79	65	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (52 neighbors)		+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+176	

Nomenclature: See separate page